

A. USEFUL LEMMAS

We first collect several technical lemmas that will be used in the proofs of Theorems 1 and 2.

Lemma 2 (Unbiasedness of the empirical objective and stochastic gradient). *For every fixed $\mathbf{x} \in \Omega$,*

$$\mathbb{E}_{\mathcal{S}_n}[\Phi_{\mathcal{S}_n}(\mathbf{x})] = \Phi_n(\mathbf{x}), \quad \mathbb{E}_{\mathcal{S}}[\Phi_{\mathcal{S}}(\mathbf{x})] = \Phi(\mathbf{x}). \quad (14)$$

Moreover, for any fixed $\mathcal{S}_n$ and $\mathbf{x}_n^k$, the stochastic gradient in (8) satisfies

$$\mathbb{E}_{\mathcal{B}}[\mathbf{g}_n^k] = \nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k), \quad (15)$$

where the expectation is taken over the mini-batch sampling.

Proof. Recall that

$$\Phi_n(\mathbf{x}) = \mu_a \left[\mathbb{E}_{\xi \sim \mathcal{D}_n} \exp \left(\frac{f(\mathbf{x}; \xi)}{\mu_w} \right) \right]^m, \quad m = \frac{\mu_w}{\mu_a}.$$

For m independent data samples $\xi_1, \dots, \xi_m \sim \mathcal{D}_n$, independence gives

$$\Phi_n(\mathbf{x}) = \mathbb{E}_{\xi_{1:m}} \left[\mu_a \exp \left(\frac{1}{\mu_w} \sum_{j=1}^m f(\mathbf{x}; \xi_j) \right) \right]. \quad (16)$$

Recall that the local empirical objective is defined as

$$\Phi_{\mathcal{S}_n}(\mathbf{x}) = \binom{Z}{m}^{-1} \sum_{\substack{\mathcal{J} \subset \mathcal{S}_n \\ |\mathcal{J}|=m}} \mu_a \exp \left[\frac{1}{\mu_w} \sum_{\xi \in \mathcal{J}} f(\mathbf{x}; \xi) \right]. \quad (17)$$

Since the data samples in $\mathcal{S}_n$ are i.i.d., for every fixed subset $\mathcal{J}$ of size m , its elements have the same joint distribution as m independent data samples $\xi_1, \dots, \xi_m \sim \mathcal{D}_n$. Therefore, averaging over all size- m subsets then yields $\mathbb{E}_{\mathcal{S}_n}[\Phi_{\mathcal{S}_n}(\mathbf{x})] = \Phi_n(\mathbf{x})$. Averaging over all agents further gives $\mathbb{E}_{\mathcal{S}}[\Phi_{\mathcal{S}}(\mathbf{x})] = \Phi(\mathbf{x})$.

We next prove the unbiasedness of the stochastic gradient. Differentiating $\Phi_{\mathcal{S}_n}$ gives

$$\nabla \Phi_{\mathcal{S}_n}(\mathbf{x}) = \binom{Z}{m}^{-1} \sum_{\substack{\mathcal{J} \subset \mathcal{S}_n \\ |\mathcal{J}|=m}} \exp \left[\frac{1}{\mu_w} \sum_{\xi \in \mathcal{J}} f(\mathbf{x}; \xi) \right] \frac{1}{m} \sum_{\xi \in \mathcal{J}} \nabla f(\mathbf{x}; \xi), \quad (18)$$

where $\mu_a/\mu_w = 1/m$ has been used.

Recall that the stochastic gradient in (8) can be written as

$$\mathbf{g}_n^k = \frac{1}{B-m+1} \sum_{b=m}^B T_b, \quad (19)$$

where

$$T_b := \exp \left[\frac{1}{\mu_w} \left(\sum_{j=1}^{m-1} f(\mathbf{x}_n^k; \xi_{n,j}^k) + f(\mathbf{x}_n^k; \xi_{n,b}^k) \right) \right] \nabla f(\mathbf{x}_n^k; \xi_{n,b}^k). \quad (20)$$

Since the mini-batch is sampled uniformly without replacement, all T_b , $b = m, \dots, B$, have the same expectation. It therefore suffices to consider T_m , which is given by

$$T_m = \exp \left[\frac{1}{\mu_w} \sum_{j=1}^m f(\mathbf{x}_n^k; \xi_{n,j}^k) \right] \nabla f(\mathbf{x}_n^k; \xi_{n,m}^k).$$

The first m data samples form a uniformly sampled size- m subset of $\mathcal{S}_n$. Therefore, taking expectation over the mini-batch sampling gives

$$\begin{aligned} \mathbb{E}_{\mathcal{B}}[T_m] &= \binom{Z}{m}^{-1} \sum_{\substack{\mathcal{J} \subset \mathcal{S}_n \\ |\mathcal{J}|=m}} \exp \left[\frac{1}{\mu_w} \sum_{\xi \in \mathcal{J}} f(\mathbf{x}_n^k; \xi) \right] \frac{1}{m} \sum_{\xi \in \mathcal{J}} \nabla f(\mathbf{x}_n^k; \xi) \\ &= \nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k), \end{aligned} \quad (21)$$

where the last equality follows directly from (18).

Finally, using (19) and the linearity of expectation,

$$\begin{aligned} \mathbb{E}_{\mathcal{B}}[\mathbf{g}_n^k] &= \frac{1}{B-m+1} \sum_{b=m}^B \mathbb{E}_{\mathcal{B}}[T_b] \\ &= \nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k). \end{aligned} \quad (22)$$

Hence, $\mathbf{g}_n^k$ is an unbiased estimator of the local empirical full gradient. $\square$

Lemma 3 (Consensus Error [15]). *Under Assumptions 2–4, suppose that all agents use the same initialization and the constant step size α is sufficiently small. Then, for any $k = 0, \dots, K$,*

$$\frac{1}{N} \sum_{n=1}^N \mathbb{E}[\|\mathbf{x}_n^k - \bar{\mathbf{x}}^k\|] \leq \sqrt{12} C_\lambda \alpha \sqrt{\sigma^2 + \delta^2}, \quad (23)$$

where $C_\lambda := \sqrt{\frac{1-\lambda}{\lambda^3}}$.

Lemma 4 (Upper Bound on Gradient Perturbation). *Under Assumptions 1 and 2, suppose that*

$$\text{Var}(\|\nabla \phi(\mathbf{x}_{\mathcal{R}}^*; \mathcal{J})\|) \leq (\sigma^*)^2,$$

then, for $Z \geq m$, the following bound holds:

$$\mathbb{E} \sup_{\mathbf{x} \in \Omega} \|\nabla \Phi_{\mathcal{S}'}(\mathbf{x}) - \nabla \Phi_{\mathcal{S}}(\mathbf{x})\| \leq \frac{(\sigma^* + LR\sqrt{d})C_1\sqrt{m}}{\sqrt{NZ}}, \quad (24)$$

where $C_1 > 0$ is the universal constant from Lemma 2 of [15].

Proof. Let $M := \lfloor \frac{Z}{m} \rfloor$. For each agent n , randomly partition mM of the Z observations into M disjoint subsets $\mathcal{J}_{n,1}, \dots, \mathcal{J}_{n,M}$. Under a uniform random partition, we have

$$\nabla \Phi_{\mathcal{S}_n}(\mathbf{x}) = \mathbb{E} \left[\frac{1}{M} \sum_{j=1}^M \nabla \left(\mu_a \exp \left[\frac{1}{\mu_w} \sum_{\xi \in \mathcal{J}_{n,j}} f(\mathbf{x}; \xi) \right] \right) \right] \quad (25)$$

Averaging (25) over all agents gives

$$\nabla \Phi_{\mathcal{S}}(\mathbf{x}) = \mathbb{E} \left[\frac{1}{NM} \sum_{n=1}^N \sum_{j=1}^M \nabla \left(\mu_a \exp \left[\frac{1}{\mu_w} \sum_{\xi \in \mathcal{J}_{n,j}} f(\mathbf{x}; \xi) \right] \right) \right]. \quad (26)$$

Apply the same construction independently to $\mathcal{S}'$, and denote the resulting subsets by $\mathcal{J}'_{n,1}, \dots, \mathcal{J}'_{n,M}$. Then,

$$\begin{aligned} & \nabla \Phi_{\mathcal{S}}(\mathbf{x}) - \nabla \Phi_{\mathcal{S}'}(\mathbf{x}) \\ &= \mathbb{E} \left[\frac{1}{NM} \sum_{n=1}^N \sum_{j=1}^M \left(\nabla \left(\mu_a \exp \left[\frac{1}{\mu_w} \sum_{\xi \in \mathcal{J}_{n,j}} f(\mathbf{x}; \xi) \right] \right) \right. \right. \\ & \quad \left. \left. - \nabla \left(\mu_a \exp \left[\frac{1}{\mu_w} \sum_{\xi \in \mathcal{J}'_{n,j}} f(\mathbf{x}; \xi) \right] \right) \right) \right]. \end{aligned} \quad (27)$$

Due to the fact that $\sup_{\mathbf{x} \in \Omega} \|\mathbb{E} G(\mathbf{x})\| \leq \mathbb{E} \sup_{\mathbf{x} \in \Omega} \|G(\mathbf{x})\|$, applying this inequality to (27) yields

$$\begin{aligned} & \mathbb{E} \sup_{\mathbf{x} \in \Omega} \|\nabla \Phi_{\mathcal{S}}(\mathbf{x}) - \nabla \Phi_{\mathcal{S}'}(\mathbf{x})\| \\ & \leq \mathbb{E} \sup_{\mathbf{x} \in \Omega} \left\| \frac{1}{NM} \sum_{n=1}^N \sum_{j=1}^M \left(\nabla \left(\mu_a \exp \left[\frac{1}{\mu_w} \sum_{\xi \in \mathcal{J}_{n,j}} f(\mathbf{x}; \xi) \right] \right) \right. \right. \\ & \quad \left. \left. - \nabla \left(\mu_a \exp \left[\frac{1}{\mu_w} \sum_{\xi \in \mathcal{J}'_{n,j}} f(\mathbf{x}; \xi) \right] \right) \right) \right\|. \end{aligned} \quad (28)$$

For each random partition, the subsets $\mathcal{J}_{n,1}, \dots, \mathcal{J}_{n,M}$ are disjoint and independent. Applying Lemma 2 in [15] gives

$$\mathbb{E} \sup_{\mathbf{x} \in \Omega} \|\nabla \Phi_{\mathcal{S}}(\mathbf{x}) - \nabla \Phi_{\mathcal{S}'}(\mathbf{x})\| \leq \frac{C_1(\sigma^* + LR\sqrt{d})\sqrt{m}}{\sqrt{NZ}}, \quad (29)$$

□

Lemma 5 (Transfer from Φ to $\mathcal{R}$). *Let $\mathbf{x}_{\mathcal{R}}^*$ denote the common minimizer of $\mathcal{R}$ and Φ . Then, for every $\mathbf{x} \in \Omega$,*

$$\mathcal{R}(\mathbf{x}) - \mathcal{R}(\mathbf{x}_{\mathcal{R}}^*) \leq \kappa_{\mathcal{R}} [\Phi(\mathbf{x}) - \Phi(\mathbf{x}_{\mathcal{R}}^*)], \quad (30)$$

where $\kappa_{\mathcal{R}} := \frac{\mu_a}{\Phi(\mathbf{x}_{\mathcal{R}}^*)}$.

Proof. Since $\mathcal{R}(\mathbf{x}) = \mu_a \log \left(\frac{\Phi(\mathbf{x})}{\mu_a} \right)$ is strictly increasing in $\Phi(\mathbf{x})$, $\mathcal{R}$ and Φ share the same minimizer. Hence, $\frac{\Phi(\mathbf{x})}{\Phi(\mathbf{x}_{\mathcal{R}}^*)} \geq 1$. Using $\log t \leq t - 1$, we obtain

$$\begin{aligned} \mathcal{R}(\mathbf{x}) - \mathcal{R}(\mathbf{x}_{\mathcal{R}}^*) &= \mu_a \log \frac{\Phi(\mathbf{x})}{\Phi(\mathbf{x}_{\mathcal{R}}^*)} \\ &\leq \mu_a \left(\frac{\Phi(\mathbf{x})}{\Phi(\mathbf{x}_{\mathcal{R}}^*)} - 1 \right) \\ &= \frac{\mu_a}{\Phi(\mathbf{x}_{\mathcal{R}}^*)} [\Phi(\mathbf{x}) - \Phi(\mathbf{x}_{\mathcal{R}}^*)], \end{aligned} \quad (31)$$

which proves (30). □

B. PROOF OF THEOREM 1

Proof. Define $\boldsymbol{\sigma}_n^k := \mathbf{g}_n^k - \nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k)$, and $\bar{\boldsymbol{\sigma}}^k := \frac{1}{N} \sum_{n=1}^N \boldsymbol{\sigma}_n^k$. Because W is doubly stochastic, averaging the update of Algorithm 1 gives

$$\bar{\mathbf{x}}^{k+1} = \bar{\mathbf{x}}^k - \alpha \left[\frac{1}{N} \sum_{n=1}^N \nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k) + \bar{\boldsymbol{\sigma}}^k \right]. \quad (32)$$

Using the equality $\|\mathbf{y} - \mathbf{z}\|^2 - \|\mathbf{x} - \mathbf{z}\|^2 = -2\langle \mathbf{x} - \mathbf{y}, \mathbf{y} - \mathbf{z} \rangle - \|\mathbf{y} - \mathbf{x}\|^2$, we have

$$\begin{aligned} & \mathbb{E} \left\| \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\|^2 - \mathbb{E} \left\| \bar{\mathbf{x}}^k - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\|^2 \\ &= -2\alpha \mathbb{E} \left\langle \frac{1}{N} \sum_{n=1}^N \nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k), \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle \\ & \quad - 2\alpha \mathbb{E} \left\langle \bar{\boldsymbol{\sigma}}^k, \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle - \mathbb{E} \left\| \bar{\mathbf{x}}^{k+1} - \bar{\mathbf{x}}^k \right\|^2. \end{aligned} \quad (33)$$

We first bound the second term on the RHS of (33). Substituting (32) gives

$$\begin{aligned} & -2\alpha \mathbb{E} \left\langle \bar{\boldsymbol{\sigma}}^k, \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle \\ &= -2\alpha \mathbb{E} \left\langle \bar{\boldsymbol{\sigma}}^k, \bar{\mathbf{x}}^k - \alpha \frac{1}{N} \sum_{n=1}^N \nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k) - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle + 2\alpha^2 \mathbb{E} \left\| \bar{\boldsymbol{\sigma}}^k \right\|^2. \end{aligned}$$

By Lemma 2, $\mathbb{E}[\bar{\boldsymbol{\sigma}}^k] = \mathbf{0}$, and $\mathbb{E}[\|\bar{\boldsymbol{\sigma}}^k\|^2] \leq \frac{\sigma^2}{N}$. The first inner product therefore vanishes, which yields

$$-2\alpha \mathbb{E} \left\langle \bar{\boldsymbol{\sigma}}^k, \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle \leq \frac{2\alpha^2 \sigma^2}{N}. \quad (34)$$

Next, by Assumption 2, we have

$$\left\| \frac{1}{N} \sum_{n=1}^N [\nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k) - \nabla \Phi_{\mathcal{S}_n}(\bar{\mathbf{x}}^k)] \right\| \leq \frac{L}{N} \sum_{n=1}^N \|\mathbf{x}_n^k - \bar{\mathbf{x}}^k\|. \quad (35)$$

For the first term on the RHS of (33), we have

$$\begin{aligned} & \left\langle \frac{1}{N} \sum_{n=1}^N \nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k), \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle \\ &= \left\langle \nabla \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^k), \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle \\ & \quad + \left\langle \frac{1}{N} \sum_{n=1}^N [\nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k) - \nabla \Phi_{\mathcal{S}_n}(\bar{\mathbf{x}}^k)], \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle \\ &\geq \left\langle \nabla \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^k), \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle \\ & \quad - \left\| \frac{1}{N} \sum_{n=1}^N [\nabla \Phi_{\mathcal{S}_n}(\mathbf{x}_n^k) - \nabla \Phi_{\mathcal{S}_n}(\bar{\mathbf{x}}^k)] \right\| \left\| \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\| \\ &\geq \left\langle \nabla \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^k), \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle - \frac{LR}{N} \sum_{n=1}^N \|\mathbf{x}_n^k - \bar{\mathbf{x}}^k\|, \end{aligned} \quad (36)$$

where the last inequality follows from (35) and Assumption 1. Furthermore, if $\alpha \leq 1/L$, according to Assumption 2, we have

$$\begin{aligned} & \left\langle \nabla \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^k), \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle + \frac{1}{2\alpha} \left\| \bar{\mathbf{x}}^{k+1} - \bar{\mathbf{x}}^k \right\|^2 \\ &= \left\langle \nabla \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^k), \bar{\mathbf{x}}^{k+1} - \bar{\mathbf{x}}^k \right\rangle + \frac{1}{2\alpha} \left\| \bar{\mathbf{x}}^{k+1} - \bar{\mathbf{x}}^k \right\|^2 \\ & \quad + \left\langle \nabla \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^k), \bar{\mathbf{x}}^k - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle \\ &\geq \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^{k+1}) - \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^k) + \left\langle \nabla \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^k), \bar{\mathbf{x}}^k - \mathbf{x}_{\Phi_{\mathcal{S}}}^* \right\rangle \\ &\geq \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^{k+1}) - \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^k) + \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^k) - \Phi_{\mathcal{S}}(\mathbf{x}_{\Phi_{\mathcal{S}}}^*) \\ &= \Phi_{\mathcal{S}}(\bar{\mathbf{x}}^{k+1}) - \Phi_{\mathcal{S}}(\mathbf{x}_{\Phi_{\mathcal{S}}}^*). \end{aligned} \quad (37)$$

Substituting (34), (36), and (37) into (33), and cancelling the step-norm term, gives

$$\begin{aligned} & \alpha \mathbb{E} \left[\Phi_S(\bar{\mathbf{x}}^{k+1}) - \Phi_S(\mathbf{x}_{\Phi_S}^*) \right] \\ & \leq \frac{1}{2} \mathbb{E} \left[\left\| \bar{\mathbf{x}}^k - \mathbf{x}_{\Phi_S}^* \right\|^2 - \left\| \bar{\mathbf{x}}^{k+1} - \mathbf{x}_{\Phi_S}^* \right\|^2 \right] \\ & \quad + \frac{\alpha^2 \sigma^2}{N} + \frac{\alpha LR}{N} \sum_{n=1}^N \mathbb{E} \left\| \mathbf{x}_n^k - \bar{\mathbf{x}}^k \right\|. \end{aligned} \quad (38)$$

Summing over $k = 0, \dots, K-1$ telescopes the squared-distance terms. Applying Lemma 3 gives

$$\begin{aligned} & \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \left[\Phi_S(\bar{\mathbf{x}}^{k+1}) - \Phi_S(\mathbf{x}_{\Phi_S}^*) \right] \\ & \leq \frac{\mathbb{E} \left\| \bar{\mathbf{x}}^0 - \mathbf{x}_{\Phi_S}^* \right\|^2}{2\alpha K} + \frac{\alpha \sigma^2}{N} + \sqrt{12} \alpha L R C_\lambda \sqrt{\sigma^2 + \delta^2}. \end{aligned} \quad (39)$$

Finally, by convexity and the definition $\hat{\mathbf{x}}^K = \frac{1}{K} \sum_{k=0}^{K-1} \bar{\mathbf{x}}^{k+1}$, we have

$$\Phi_S(\hat{\mathbf{x}}^K) \leq \frac{1}{K} \sum_{k=0}^{K-1} \Phi_S(\bar{\mathbf{x}}^{k+1}).$$

Choosing $\alpha = \Theta(K^{-1/2})$ in (39) proves Theorem 1. $\square$

C. PROOF OF THEOREM 2

Proof. Let S' be an independent copy of S , and let $\hat{\mathbf{x}}'^K$ denote the output of Algorithm 1 trained on S' . According to Lemma 1,

$$\mathbb{E} \left[\Phi(\hat{\mathbf{x}}^K) - \Phi_S(\hat{\mathbf{x}}^K) \right] \leq \mathbb{E} \left[\Phi_S(\hat{\mathbf{x}}'^K) - \Phi_S(\mathbf{x}_{\Phi_S}^*) \right]. \quad (40)$$

Thus, it remains to bound the empirical suboptimality of the trajectory trained on S' when evaluated on Φ_S .

Let $\{\mathbf{x}_n'^k\}$ denote this trajectory. Define $\boldsymbol{\sigma}_n'^k := \mathbf{g}_n'^k - \nabla \Phi_{S'_n}(\mathbf{x}_n'^k)$, and $\boldsymbol{\zeta}_n^k := \nabla \Phi_{S'_n}(\mathbf{x}_n'^k) - \nabla \Phi_{S_n}(\mathbf{x}_n'^k)$. The stochastic gradient can then be written as

$$\mathbf{g}_n'^k = \nabla \Phi_{S_n}(\mathbf{x}_n'^k) + \boldsymbol{\sigma}_n'^k + \boldsymbol{\zeta}_n^k. \quad (41)$$

The second term is conditionally zero mean, while the third term is the dataset-induced gradient perturbation.

Define $\bar{\boldsymbol{\sigma}}'^k := \frac{1}{N} \sum_{n=1}^N \boldsymbol{\sigma}_n'^k$, and $\bar{\boldsymbol{\zeta}}^k := \frac{1}{N} \sum_{n=1}^N \boldsymbol{\zeta}_n^k$. Double stochasticity of W yields

$$\bar{\mathbf{x}}'^{k+1} = \bar{\mathbf{x}}'^k - \alpha \left[\frac{1}{N} \sum_{n=1}^N \nabla \Phi_{S_n}(\mathbf{x}_n'^k) + \bar{\boldsymbol{\sigma}}'^k + \bar{\boldsymbol{\zeta}}^k \right]. \quad (42)$$

Applying the same argument as in the proof of Theorem 1 gives

$$\begin{aligned} & \mathbb{E} \left\| \bar{\mathbf{x}}'^{k+1} - \mathbf{x}_{\Phi_S}^* \right\|^2 - \mathbb{E} \left\| \bar{\mathbf{x}}'^k - \mathbf{x}_{\Phi_S}^* \right\|^2 \\ & = -2\alpha \mathbb{E} \left\langle \frac{1}{N} \sum_{n=1}^N \nabla \Phi_{S_n}(\mathbf{x}_n'^k), \bar{\mathbf{x}}'^{k+1} - \mathbf{x}_{\Phi_S}^* \right\rangle \\ & \quad - 2\alpha \mathbb{E} \left\langle \bar{\boldsymbol{\sigma}}'^k, \bar{\mathbf{x}}'^{k+1} - \mathbf{x}_{\Phi_S}^* \right\rangle \\ & \quad - 2\alpha \mathbb{E} \left\langle \bar{\boldsymbol{\zeta}}^k, \bar{\mathbf{x}}'^{k+1} - \mathbf{x}_{\Phi_S}^* \right\rangle - \mathbb{E} \left\| \bar{\mathbf{x}}'^{k+1} - \bar{\mathbf{x}}'^k \right\|^2. \end{aligned} \quad (43)$$

The stochastic-noise term is bounded exactly as in (34):

$$-2\alpha \mathbb{E} \left\langle \bar{\boldsymbol{\sigma}}'^k, \bar{\mathbf{x}}'^{k+1} - \mathbf{x}_{\Phi_S}^* \right\rangle \leq \frac{2\alpha^2 \sigma^2}{N}. \quad (44)$$

Similarly, the first and fourth terms on the RHS of (43) satisfy

$$\begin{aligned} & \left\langle \frac{1}{N} \sum_{n=1}^N \nabla \Phi_{S_n}(\mathbf{x}_n'^k), \bar{\mathbf{x}}'^{k+1} - \mathbf{x}_{\Phi_S}^* \right\rangle + \frac{1}{2\alpha} \left\| \bar{\mathbf{x}}'^{k+1} - \bar{\mathbf{x}}'^k \right\|^2 \\ & \geq \Phi_S(\bar{\mathbf{x}}'^{k+1}) - \Phi_S(\mathbf{x}_{\Phi_S}^*) - \frac{LR}{N} \sum_{n=1}^N \left\| \mathbf{x}_n'^k - \bar{\mathbf{x}}'^k \right\|. \end{aligned} \quad (45)$$

We next control the additional dataset perturbation.

$$\begin{aligned} \left\| \bar{\boldsymbol{\zeta}}^k \right\| & \leq \left\| \nabla \Phi_{S'}(\bar{\mathbf{x}}'^k) - \nabla \Phi_S(\bar{\mathbf{x}}'^k) \right\| \\ & \quad + \frac{1}{N} \sum_{n=1}^N \left\| \nabla \Phi_{S'_n}(\mathbf{x}_n'^k) - \nabla \Phi_{S'_n}(\bar{\mathbf{x}}'^k) \right\| \\ & \quad + \frac{1}{N} \sum_{n=1}^N \left\| \nabla \Phi_{S_n}(\mathbf{x}_n'^k) - \nabla \Phi_{S_n}(\bar{\mathbf{x}}'^k) \right\| \\ & \leq \sup_{\mathbf{x} \in \Omega} \left\| \nabla \Phi_{S'}(\mathbf{x}) - \nabla \Phi_S(\mathbf{x}) \right\| + \frac{2L}{N} \sum_{n=1}^N \left\| \mathbf{x}_n'^k - \bar{\mathbf{x}}'^k \right\|. \end{aligned} \quad (46)$$

Taking expectations in (46) and using the diameter bound in Assumption 1 give

$$\begin{aligned} & -2\alpha \mathbb{E} \left\langle \bar{\boldsymbol{\zeta}}^k, \bar{\mathbf{x}}'^{k+1} - \mathbf{x}_{\Phi_S}^* \right\rangle \\ & \leq 2\alpha R \mathbb{E} \left\| \bar{\boldsymbol{\zeta}}^k \right\| \\ & \leq 2\alpha R \mathbb{E} \sup_{\mathbf{x} \in \Omega} \left\| \nabla \Phi_{S'}(\mathbf{x}) - \nabla \Phi_S(\mathbf{x}) \right\| \\ & \quad + \frac{4\alpha LR}{N} \sum_{n=1}^N \mathbb{E} \left\| \mathbf{x}_n'^k - \bar{\mathbf{x}}'^k \right\|. \end{aligned} \quad (47)$$

Combining (44), (45), and (47) with (43) gives

$$\begin{aligned} & \alpha \mathbb{E} \left[\Phi_S(\bar{\mathbf{x}}'^{k+1}) - \Phi_S(\mathbf{x}_{\Phi_S}^*) \right] \\ & \leq \frac{1}{2} \mathbb{E} \left[\left\| \bar{\mathbf{x}}'^k - \mathbf{x}_{\Phi_S}^* \right\|^2 - \left\| \bar{\mathbf{x}}'^{k+1} - \mathbf{x}_{\Phi_S}^* \right\|^2 \right] \\ & \quad + \frac{\alpha^2 \sigma^2}{N} + \alpha R \mathbb{E} \sup_{\mathbf{x} \in \Omega} \left\| \nabla \Phi_{S'}(\mathbf{x}) - \nabla \Phi_S(\mathbf{x}) \right\| \\ & \quad + \frac{3\alpha LR}{N} \sum_{n=1}^N \mathbb{E} \left\| \mathbf{x}_n'^k - \bar{\mathbf{x}}'^k \right\|. \end{aligned} \quad (48)$$

Summing over $k = 0, \dots, K-1$, using convexity of Φ_S , and applying Lemmas 3 and 4 to the trajectory trained on S' yield

$$\begin{aligned} & \mathbb{E} \left[\Phi_S(\hat{\mathbf{x}}'^K) - \Phi_S(\mathbf{x}_{\Phi_S}^*) \right] \\ & \leq \frac{\mathbb{E} \left\| \bar{\mathbf{x}}'^0 - \mathbf{x}_{\Phi_S}^* \right\|^2}{2\alpha K} + \frac{\alpha \sigma^2}{N} \\ & \quad + 6\sqrt{3} \alpha L R C_\lambda \sqrt{\sigma^2 + \delta^2} + \frac{C_1 R (\sigma^* + LR\sqrt{d}) \sqrt{m}}{\sqrt{N} Z}. \end{aligned} \quad (49)$$

Combining this bound with Lemma 1 and choosing $\alpha = \Theta(K^{-1/2})$ proves Theorem 2. $\square$

D. PROOF OF THEOREM 3

Proof. Applying Lemma 5 at $\mathbf{x} = \widehat{\mathbf{x}}^K$ gives

$$\mathbb{E}[\mathcal{R}(\widehat{\mathbf{x}}^K) - \mathcal{R}(\mathbf{x}_{\mathcal{R}}^*)] \leq \kappa_{\mathcal{R}} \mathbb{E}[\Phi(\widehat{\mathbf{x}}^K) - \Phi(\mathbf{x}_{\Phi}^*)], \quad (50)$$

where $\mathbf{x}_{\mathcal{R}}^* = \mathbf{x}_{\Phi}^*$ since $\mathcal{R}$ and Φ share the same minimizer.

Inserting the empirical objective Φ_S and its minimizer $\mathbf{x}_{\Phi_S}^*$, we have

$$\begin{aligned} & \mathbb{E}[\Phi(\widehat{\mathbf{x}}^K) - \Phi(\mathbf{x}_{\Phi}^*)] \\ &= \mathbb{E}[\Phi(\widehat{\mathbf{x}}^K) - \Phi_S(\widehat{\mathbf{x}}^K)] + \mathbb{E}[\Phi_S(\widehat{\mathbf{x}}^K) - \Phi_S(\mathbf{x}_{\Phi_S}^*)] \\ & \quad + \mathbb{E}[\Phi_S(\mathbf{x}_{\Phi_S}^*) - \Phi(\mathbf{x}_{\Phi}^*)]. \end{aligned} \quad (51)$$

Since $\mathbf{x}_{\Phi_S}^*$ minimizes Φ_S ,

$$\Phi_S(\mathbf{x}_{\Phi_S}^*) \leq \Phi_S(\mathbf{x}_{\Phi}^*).$$

Together with $\mathbb{E}\Phi_S(\mathbf{x}_{\Phi}^*) = \Phi(\mathbf{x}_{\Phi}^*)$, this gives

$$\mathbb{E}[\Phi_S(\mathbf{x}_{\Phi_S}^*) - \Phi(\mathbf{x}_{\Phi}^*)] \leq 0.$$

Therefore,

$$\begin{aligned} & \mathbb{E}[\Phi(\widehat{\mathbf{x}}^K) - \Phi(\mathbf{x}_{\Phi}^*)] \\ & \leq \mathbb{E}[\Phi(\widehat{\mathbf{x}}^K) - \Phi_S(\widehat{\mathbf{x}}^K)] + \mathbb{E}[\Phi_S(\widehat{\mathbf{x}}^K) - \Phi_S(\mathbf{x}_{\Phi_S}^*)]. \end{aligned} \quad (52)$$

Applying Theorems 2 and 1 to the two terms on the RHS of (52), respectively, and then using (50), we obtain

$$\mathbb{E}[\mathcal{R}(\widehat{\mathbf{x}}^K) - \mathcal{R}(\mathbf{x}_{\mathcal{R}}^*)] = O\left(\frac{1}{\sqrt{NZ}}\right) + O\left(\frac{1}{\sqrt{K}}\right), \quad (53)$$

which proves Theorem 3. $\square$